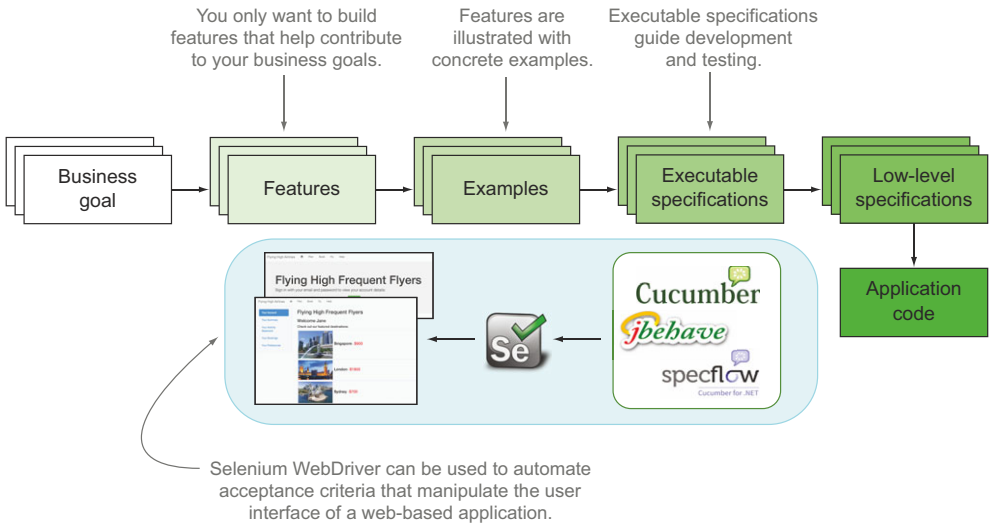


Figure 8.1 In this chapter you'll learn how to automate acceptance criteria in order to exercise the UI of web-based applications.

the overall user experience. The screenshots from automated web tests can be a valuable aid for testers, and they're also a great way to provide illustrated documentation describing how the application behaves.

We'll look at automated web testing from several perspectives:

- *Automated web tests* are very effective at testing user interactions with the UI, and for illustrating end-to-end user interactions with the application. But if they're badly designed or used to test things that would be better tested by non-UI tests, web tests can become a maintenance liability.
- *Selenium WebDriver* is a popular open source library that can be used to write automated web tests in a number of languages, such as Java, Groovy, Ruby, C#, and Python.
- Selenium WebDriver provides good support for *Page Objects*. The Page Objects design pattern can to help make automated web tests more readable and easier to maintain.
- Many open source libraries build upon and extend Selenium WebDriver to make writing automated web tests easier and more convenient.

Although we'll focus on web testing in this chapter, many of the tools and approaches discussed here also apply to other types of user interfaces. For example, mobile apps can be tested effectively using the toolset we'll discuss here by using Appium (http://appium.io/), a WebDriver-based automation library for mobile apps, and the Page Objects pattern is applicable for any type of GUI.

To write effective automated web tests, you need to know not only how to automate web tests well, but also when you should and shouldn't automate scenarios with web tests.